

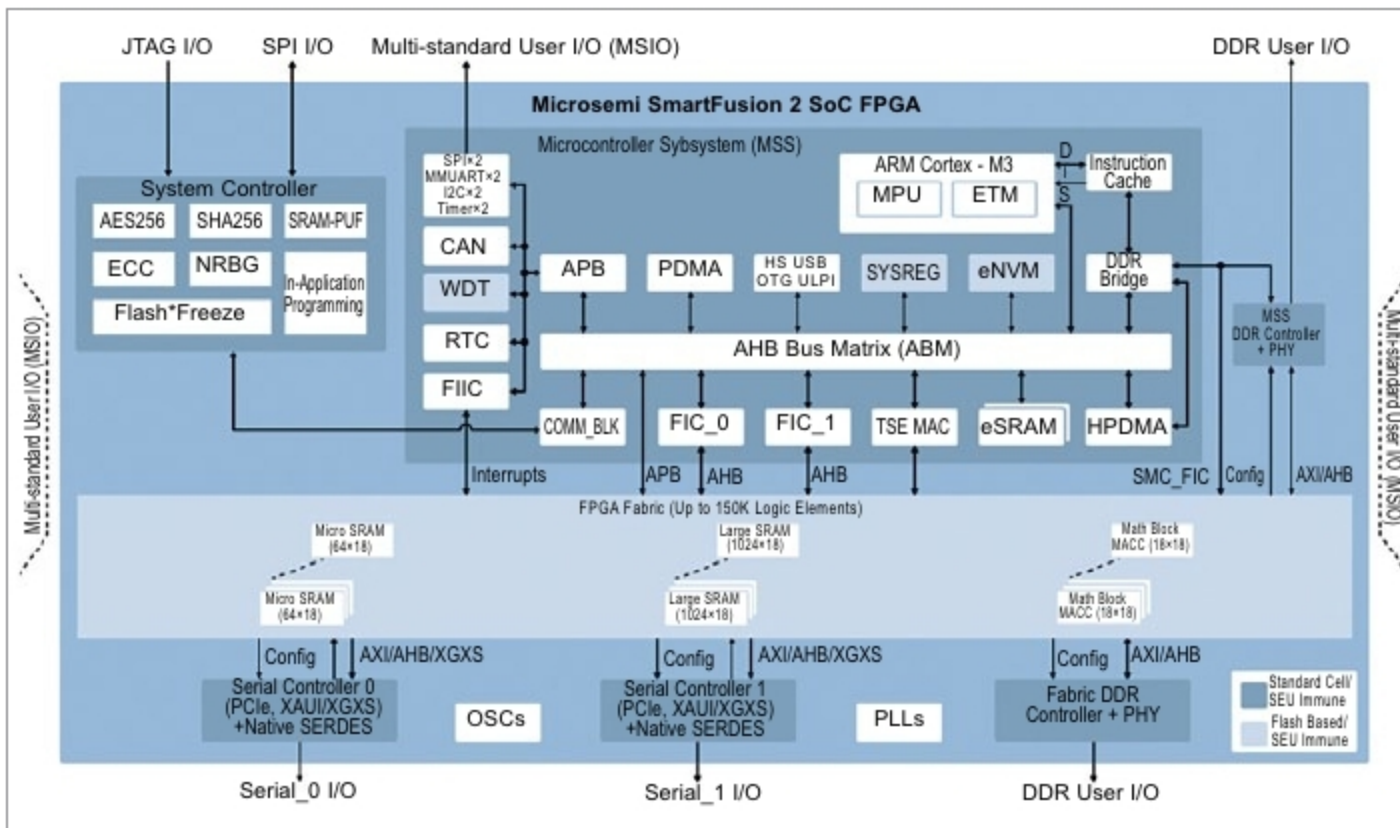


Fig. 2: SmartFusion2 SoC FPGA architecture

and instruction cache with on-chip eSRAM and eNVM, and a complete microcontroller (MCU) sub-system with extensive peripherals, including CAN, TSE and USB.

The architecture includes the following:

1. Up to 16 high-speed transceiver lanes
2. PCIe generation 2, XAUI/XGXS +, generic (EPCS) mode at 3.2G
3. Up to 150,000 LEs, 5Mbit SRAM, 4Mbit eNVM
4. Hard 667mbps DDR2/3 controllers
5. Integrated DSP processing blocks
6. Power as low as 7mW standby (typical)
7. DPA hardened, AES256, SHA256
8. On-demand NVM data integrity check
9. SEU-protected/tolerant memories: eSRAMs, DDR Bridges

Achronix FPGA

Speedcore is the optimal hardware accelerator computation and communication infrastructure in data centres. Enterprises can no longer keep pace with exponential data growth rates, changing security and software virtualisation requirements. Traditional multi-core CPUs and SoCs need programmable hardware accelerators that pre-process and offload data to increase their compute performance.

In 2018, this centralisation is reaching its limit. Volume of data that will be needed to drive the next wave of applications is beginning to force a change in direction. Currently, only 10 per cent of enterprise-generated data is created and processed outside centralised data centres. Industry analyst Gartner predicts this figure will reach 50 per cent by 2022. This necessary reversal is driven by the shift towards hyper-connected cyber-physical systems, enabled by the arrival of technologies such as 5G

wireless communications and a new wave of application-focused computing hardware.

Internet of Things (IoT) devices generate small individual amounts of data that are most efficiently aggregated and processed in large data centres. But as IoT devices and, increasingly, cyber-physical en-masse systems come to rely on the ability to interpret much more substantial data streams, the center of gravity needs to shift towards the edge. The automotive sector adopts changes that will be seen across many sectors.

Speedcore is the optimal solution for high-volume applications and offers significant advantages such as lower power, higher interface performance, lower system cost, higher system reliability and yields, and process technology to evaluate Speedcore. It is designed for computer and network acceleration applications, and is based on the same high-performance architecture.

Speedcore eFPGA products are fully supported by Achronix's robust and proven ACE design tools. With Speedcore, customers specify opti-

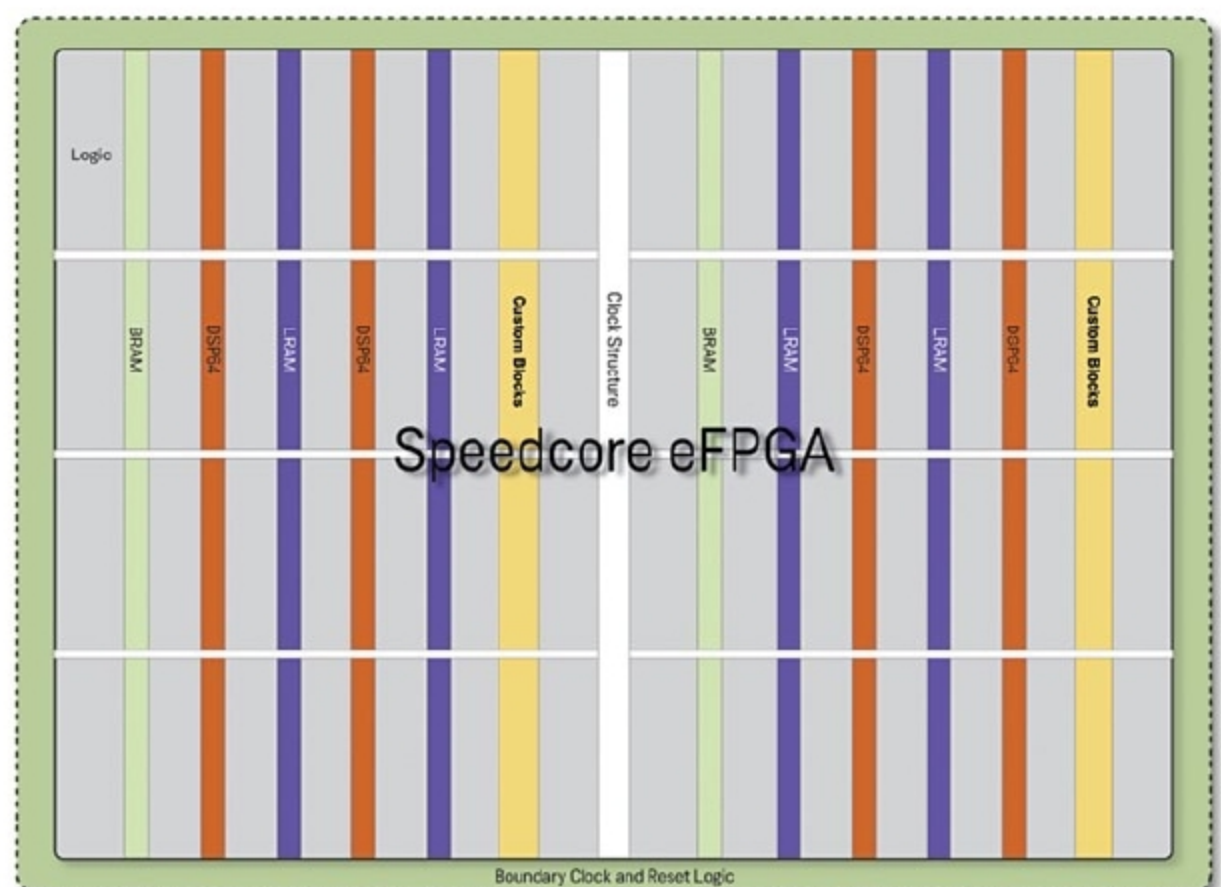


Fig. 3: Speedcore eFPGA